

User Experience Design Process

Module 3

116U01E734

User Experience Design

July-Dec 2025

User Research and Journey

- Defining the UX Design Process and its Methodology, Understanding user needs, Flow chart, Six stages of UX Design process.
- The four quadrants of empathy map, emotional mapping using an empathy map, Design Thinking, Wicked problems, Ideation.

Defining UX Design Process

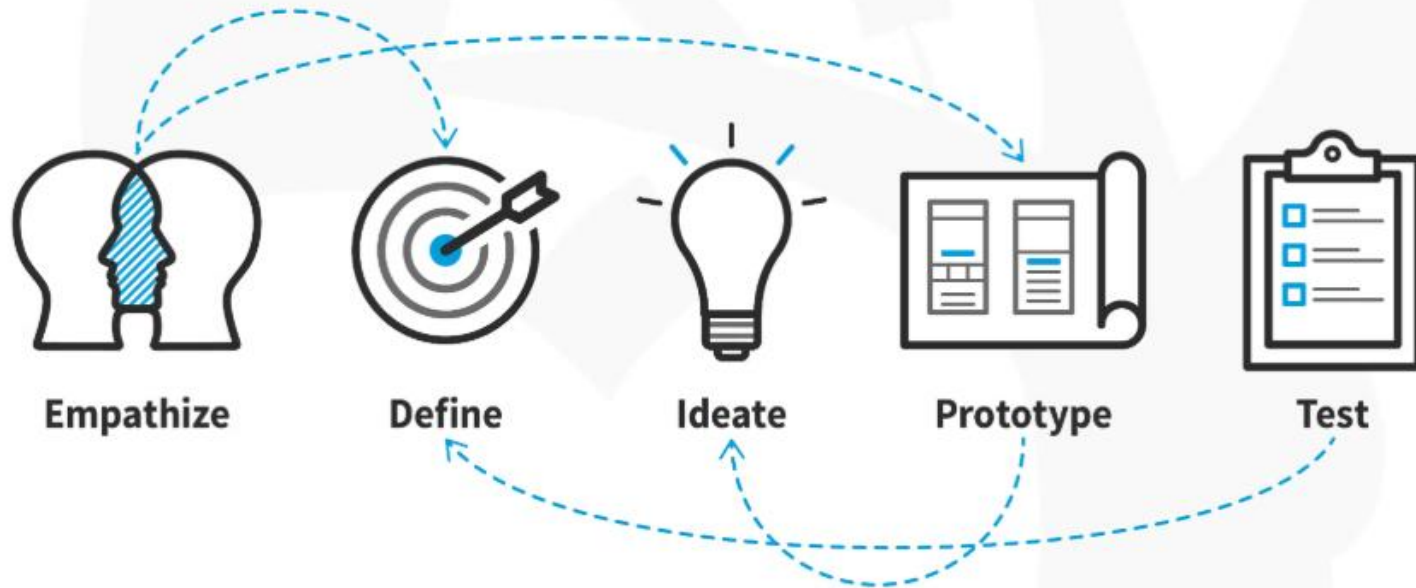
- The following steps can take you all the way from an idea to a product launch:
 - Research
 - Define the user problem
 - Design



Defining UX Design Process



Design Thinking: A 5-Stage Process



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Defining UX Design Process

- **Research and define problem**
 - **Understand the users' needs**
 - **User Centric Design**
 - Involves speaking to or observing real users (or people who represent your target users) to figure out what problems they have and what they desire and require from a solution
 - **Qualitative**
 - which involves learning about users' thoughts, feelings and opinions
 - **Quantitative**
 - measurable data such as how many times a user clicks on a certain button or how long it takes them to complete a particular task

Defining UX Design Process

- **Define the user problem**
 - a short summary of the problem you're going to design a solution for
 - To understand approach to solve the problem, problem statement is divided into a series of “How Might We?” questions.

Defining UX Design Process

- **Design**
 - Come up with potential solutions to your user problem
 - Choose one solution that is most feasible and most likely to meet the users' needs bringing to life by designing
 - Might include sketching out initial ideas for how the new feature will work considering things like:
 - what functionality it will provide
 - where it'll sit in the app
 - how the user will get to it.

Defining UX Design Process

- **Design**
 - After the initial ideas sketched out the information about architecture of the new feature (i.e. how it will be structured and organised in terms of content and information) is consider
 - User flows, a chart or diagram which depicts the path a user will take to complete a certain task is mapped out
 - For new features, create user flows to determine how the user will complete tasks is created,
 - Create wireframes (very low-fidelity wireframes) or digitally using a variety of wireframing tools are created to further firm up the design of your new feature.

Defining UX Design Process

- **Prototype**
 - After laying out the blueprint for new product or feature, the details of the design are fine tuned by creating a more lifelike model of how the product will look and function.
 - A simulation that shows how users will interact with the finished product.
 - Can be interactive and clickable, allowing stakeholders (and sample users) to interact with them as if they're a live product.

Defining UX Design Process

- **Prototype**
 - The purpose is to give something to the user to test before spending time and money in developing design into a living and breathing product.
 - Enables to make sure that the solution proposed is user-friendly and accessible.
 - Ensures that users are able to interact with it in the way originally intended.

Defining UX Design Process

- **Validate**
 - Test designs to make sure they're effectively solving the user problem and that they're a joy to interact with.
 - This step in the process requires UX testing (or usability testing) on real or representative users.
 - Gives the chance to identify problems with designs and fix them before development, ultimately saving time, money and disgruntled users.

Defining UX Design Process

- **Validate**
 - UX testing allows to validate or invalidate designs, determining whether to move forward to the development stage or return to the design stage to make adjustments..

Understanding Users needs in UX Design Process

- **Need of understanding:**
 - Prevents wasted resources
 - Improves satisfaction and adoption
 - Creates loyalty & trust

Understanding Users needs in UX Design Process

- **Need of understanding:**
 - Prevents wasted resources
 - Ensuring that development efforts are focused on creating solutions that are actually needed and desired by users, rather than building features that are unused or irrelevant.

Understanding Users needs in UX Design Process

- **Prevents wasted resources:**
- **Avoiding Unnecessary Features and Development:**
 - **Targeted Development:**
 - By understanding user needs, designers can identify the most important features and functionalities that users actually require.
 - This prevents the development of unnecessary features that consume resources without adding value.
 - **Reduced Redesign Cycles:**
 - Thorough user research helps identify potential usability issues early in the design process.
 - Addressing these issues early on reduces the need for costly redesigns and rework later in the development cycle.

Thus Focus on Core Functionality

Understanding Users needs in UX Design Process

- **Prevents wasted resources:**
- **Minimizing User Errors and Support Costs:**
 - **Intuitive Interfaces:**
 - Understanding user behavior and pain points allows designers to create intuitive and easy-to-use interfaces, reducing the likelihood of user errors and the need for extensive user support.
 - **Reduced Support Tickets:**
 - Fewer user errors and frustrations translate into fewer support tickets and reduced costs associated with customer service and technical support.

Understanding Users needs in UX Design Process

- **Prevents wasted resources:**
- **Improving User Satisfaction and Retention:**
 - **Increased Engagement:**
 - When users find a product or service that meets their needs and solves their problems, they are more likely to be engaged and satisfied, leading to higher user retention rates.
 - **Positive Word-of-Mouth:**
 - Satisfied users are more likely to recommend the product or service to others, leading to organic growth and reduced marketing costs.

Understanding Users needs in UX Design Process

- **Prevents wasted resources:**
- **Optimizing Development Processes:**
 - **Lean UX Practices:**
 - Lean UX methodologies emphasize iterative development and continuous user feedback.
 - By understanding user needs, teams can focus on delivering value in each iteration, minimizing waste and maximizing efficiency.

Data-Driven Decisions

Understanding Users needs in UX Design Process

- **Prevents wasted resources:**
- **Promoting Sustainable UX:**
 - **Reduced Environmental Impact:**
 - Understanding user behavior can lead to more efficient designs that consume less energy and resources.
 - Optimizing image sizes, reducing data transfer, and designing for less cognitive load can all contribute to a more sustainable user experience
 - **Longer Product Lifespan:**
 - Designing for durability, repairability, and upgradability can extend the lifespan of digital products, reducing e-waste and promoting a more sustainable approach,

Understanding Users needs in UX Design Process

- **Need of understanding:**
 - **Improves satisfaction and adoption**
 - By tailoring the design to meet user expectations and address their pain points, designers can create more intuitive, efficient, and enjoyable experiences, leading to higher satisfaction and increased adoption rates.

Understanding Users needs in UX Design Process

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Understanding Users needs in UX Design Process

- **Need of understanding:**
- **Improves satisfaction and adoption**
- **Increased User Satisfaction:**
 - **Personalized Experiences:**
 - When designers understand user needs, they can create personalized experiences that resonate with individual users.
 - Involve tailoring content, features, or even the overall interface to better suit specific user preferences and workflows.
 - **Reduced Frustration:**
 - By addressing user pain points and designing intuitive interfaces, UX designers can minimize user frustration and create a more seamless experience.
 - Particularly important for complex tasks or products where users might otherwise struggle to achieve their goals.

Understanding Users needs in UX Design Process

- **Need of understanding:**
- **Improves satisfaction and adoption**
- **Increased User Satisfaction:**
 - **Enhanced Engagement:**
 - Products that meet user needs are more likely to engage users, leading to increased time spent on the platform and a higher likelihood of return visits.
 - **Positive Feedback:**
 - Users are more likely to provide positive feedback and recommendations for products that meet their needs effectively.

Understanding Users needs in UX Design Process

- **Need of understanding:**
- **Improved Product Adoption:**
 - **Intuitive Interfaces:**
 - When UX designers understand user needs, they can create interfaces that are easy to learn and use.
 - This reduces the learning curve and encourages users to adopt the product quickly.
 - **Reduced Barriers to Entry:**
 - By addressing user pain points and providing clear guidance, UX designers can lower the barriers to entry for new users, making it easier for them to start using the product.

Understanding Users needs in UX Design Process

- **Need of understanding:**
- **Improved Product Adoption:**
 - **Intuitive Interfaces:**
 - A well-designed user experience can create a positive first impression, encouraging users to explore the product further and ultimately adopt it.
 - **Increased Efficiency:**
 - When users can easily complete tasks and achieve their goals, they are more likely to adopt the product as a regular part of their workflow.

Understanding Users needs in UX Design Process

- **Need of understanding:**
- **Improved Product Adoption:**
 - **Higher Conversion Rates:**
 - Understanding user needs can lead to optimized user journeys, reducing friction and increasing conversion rates.
 - **Reduced Churn:**
 - A positive user experience can foster loyalty and reduce churn (users abandoning the product), as users are more likely to stick with a product that meets their needs.

Understanding Users needs in UX Design Process

- **Need of understanding:**
 - Creates loyalty & trust
 - By improving the user experience, user trust encourages customer loyalty.
 - Satisfied customers are loyal customers who will stick to the system and continue using it

Understanding Users needs in UX Design Process

- **Ways to build user trust using UX Design**
 - **Map out the customer journey to audit the whole UX**
 - the tracking of a customer's progression through your product.
 - **Use mental models to create an intuitive design**
 - Mental models are patterns that shape how we make sense of the world. When users first encounter a product, these pre-existing models determine how they interact with the product

Understanding Users needs in UX Design Process

- **Ways to build user trust using UX Design**
 - **Perform early usability testing to adapt to customer needs**
 - Helps UX designers understand customer actions and needs, a UX research method that measures how easy it is for users to use product
 - **Provide testimonials to build user trust**
 - People trust other people.
 - The quickest way to build trust is to show that others also find system trustworthy.
 - They put potential customers at ease by informing them that system have been trusted by others in the past who were satisfied with product or service

Understanding Users needs in UX Design Process

- **Ways to build user trust using UX Design**
 - **Create a positive first impression to gain users' trust**
 - The impression when a user first launches will forever impact how they judge product
 - **Replace empty states with personalized content**
 - An empty state is a screen inside product that is practically blank. These screens are common when users newly sign up to a platform as they're yet to fill in any data.
 - Empty states are an example of poor UX.
 - Left alone, they may make users freeze, confused about where to start or how to proceed.
 - Rather than leave screens blank, educate the user.
 - Replace (or fill) them with demo content or instructions that illustrate their purpose.

Understanding Users needs in UX Design Process

- **Types of User Needs**
 - Functional needs (features, performance)
 - Emotional needs (feelings, comfort, trust)
 - Social needs (interaction, belonging)

Functional Needs

- directly impacts the product's usefulness and its ability to solve user problem
- Clearly defined, designers can create intuitive user flows and interfaces that make it easy for users to complete tasks.
- **features, performance**

Emotional Needs

- UX design refer to how a product makes users **feel** on a deeper, psychological level
- The difference between a product that is merely usable and one that is **loved** and cherished.

Feelings, comfort, trust

- **Creates Brand Loyalty**
- **Increases Engagement:**
 - Products that are designed to be joyful, surprising, or rewarding encourage users to spend more time interacting with them
 - Achieved through **gamification** (using game-like elements like badges or streaks) and thoughtful **microinteractions** (small animations that provide feedback and delight, like the "like" button animation on Instagram).
- **Builds Trust:**
 - By designing an experience that feels secure and predictable, you can make users feel at ease.
 - Clear, reassuring language on error pages or during a checkout process can alleviate anxiety and build trust.

Emotional Needs

- **Norman's Three Levels of Emotional Design:**

1. Visceral:

This level is about The immediate, subconscious reaction to a product's appearance.

The first impressions and aesthetics.

- A sleek design, a vibrant color palette, or a beautiful typography choice can evoke a strong, positive visceral response.
- This level is largely about how the product **looks**.

Emotional Needs

- Norman's Three Levels of Emotional Design:

2. Behavioral:

This level is about the user's emotions during the interaction.

- **Usability and functionality**
 - When a product is intuitive, efficient, and easy to use, it creates feelings of satisfaction and competence.
 - Conversely, a clunky, confusing interface leads to frustration and anxiety. This level is about how the product **works**.

Emotional Needs

- **Norman's Three Levels of Emotional Design:**

3. Reflective:

This is the highest level of emotional design, where the user forms a conscious, long-term relationship with the product.

- It's about the meaning the user associates with the product, their personal memories, and what the product says about them.
- **storytelling, personalization, and a sense of community** come into play.
- This level is about how the product makes the user **feel and think about themselves**.

Social Needs

- Refer to the human desire for connection, belonging, and interaction with others

Interaction, belonging

– Fosters Community and Connection:

- Humans are inherently social creatures.
- Products that allow users to connect, share, and collaborate with others can build strong, loyal communities.

– Increases Engagement and Retention:

- When a product fulfills a social need, users are more likely to return.
- The desire to see what friends are doing, to get feedback from peers, or to collaborate on a project creates a powerful reason for a user to keep coming back, even if a new, more feature-rich competitor emerges

– Enhances Perceived Value:

Understanding Users needs in UX Design Process

- **Prioritize Needs**
 - MoSCoW method (Must, Should, Could, Won't)
 - Balancing business goals & user needs

Understanding Users needs in UX Design Process

- **Best Practices**
 - Continuous feedback loops
 - Diverse participant groups
 - Test early, test often

Defining UX Design Process

- **Launch**

Understanding User Needs

- Why It Matters in the Industry
 - Companies like **Google**, **Airbnb**, **Spotify**, and **Zomato** invest in **UX research** to ensure product-market fit.
 - Startups use **lean UX** to pivot ideas quickly based on user feedback.
 - UX insights directly impact **retention**, **engagement**, and **conversions**.

Example

- **Spotify** discovered that many users didn't understand how to create playlists.
- This led to reworking their onboarding process and UI flow — a clear example of how **understanding user pain points** improves UX.



Spotify Playlist Creation: UX Evolution (2020–2025)

Year	Milestone	Details
2020	UX Research Discovery	Spotify found many users didn't understand how to create or manage playlists. This was especially true for new users and those unfamiliar with streaming apps.
2021–2022	Interface Improvements	Spotify redesigned its mobile UI to make playlist creation more intuitive. The “Create Playlist” button became more prominent, and onboarding flows were enhanced.
2023	Reinforcement Learning for Playlist Generation	Spotify Research introduced a simulation-based reinforcement learning model to automatically generate playlists tailored to user preferences. This model used a simulated environment to optimize playlist satisfaction without risking real user experience.



Spotify Playlist Creation: UX Evolution (2020–2025)

Year	Milestone	Details
2024	Demand Drivers Study	A major study published in Marketing Science analysed what drives playlist followers. It found that featuring playlists on the Search Page increased daily followers by 0.95%, while adding songs by major label artists increased followers by 0.45%.
2025	Personalization & Promotion	Spotify continues to refine its algorithms and UI to promote user-generated playlists and improve discoverability. Featuring playlists prominently remains a key strategy to guide user attention and engagement.

Comparison

Feature	Spotify (2025)	Apple Music (2025)	YouTube Music (2025)
Create Playlist Button	+ Create Playlist	New Playlist	New Playlist
Custom Cover Art	✓ Yes	✓ Yes	✓ Yes
Collaborative Editing	✓ Real-time	✗ Not supported	✗ Not supported
Smart Recommendations	✓ Based on listening	✓ Personalized picks	✓ Mood-based sorting
Custom Transitions	✓ Premium feature	✓ AutoMix	✗ Not available
Playlist Analytics	✗ Limited	✗ Not available	✓ Creator dashboard

UX Design Process

- The four quadrants of empathy map, emotional mapping using an empathy map, Design Thinking, Wicked problems, Ideation.

User Design Process: Empathy map

- A square divided into four quadrants with the user or client in the middle



UX Design Process

- Empathy map:
 - A collaborative visualization used to articulate and gain the deeper understanding of end users
 - Used throughout any UX process to establish common ground among team members and to understand and prioritize user needs
 - **In user-centered design**, empathy maps are best used from the very **beginning of the design process**.
- An efficient tool used by designers to:
 - understand users' behavior
 - visually communicate those findings to colleagues
 - unit the team under one shared understanding of the user

An attempt to limit miscommunication and misunderstanding about target audiences, including customers and users

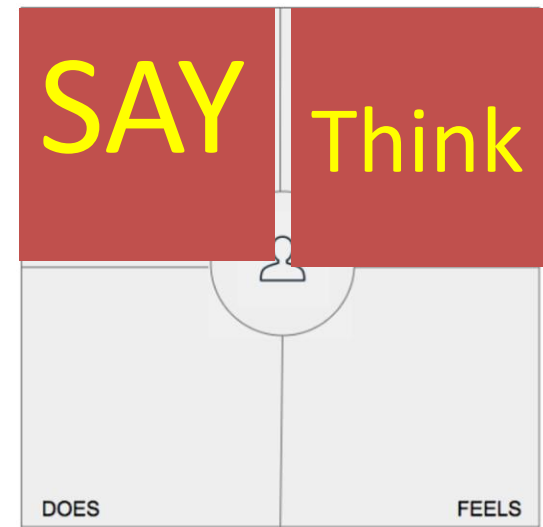
UX Design Process

- **Empathy map:**
- During user research, a lot about your users is learnt by analyzing
 - what they say
 - What they do
 - through more subtle clues like body language and facial expressions.

An empathy map visualizes all these findings and present the data gathered about users during the research phase in a condensed, easily digestible format

Empathy Maps

- The four empathy map quadrants look at user **says**, **thinks**, **feels**, and **does**.
- **Says:**
 - This section contains direct quotes from the user that have been gathered from the research phase or previous data.
- **Thinks:**
 - it is more focused on what a user is thinking and doesn't choose to say out loud.
 - Use qualitative research to ask what matters to the user and what is on their mind.
 - Looking at why they might be hesitant to share their thoughts out loud can reveal even further insight into the user and how they relate to the product or experience at hand



Empathy Maps

- **Feels:**

- This section addresses the user's emotional state and answers questions like “What is the user feeling during this product experience?” and “What worries or excites the user?”

An easy way to organize this information is to list the emotions being elicited followed by a short description of what is making the user feel this way. For example; “Overwhelmed—too many decisions to make,” or, “Anxious—doesn't want to waste their time.



Empathy Maps

- **Does:**
 - captures what the user physically does and how they do it
 - what actions does the user take and how do they take them?



Empathy Maps

- **Merits:**
- Helps us understand our user's needs and goals in a more accurately and organized way.
- Allows to see the complexity of their needs at times (for example, when they say one thing and do another), and find solutions that meet them where they are.
- Be read and understood quite easily, making them a great tool for communicating information about the user to other members of the design team.

Empathy Maps

- **Merits:**
- It's critical to help others on your team and in your company to cultivate a deeper understanding of the user's behaviors and empathy for their needs.
- Ensures that users' needs will take centerstage in design decision making, since everyone contributing to the product's development can work to serve the same set of personas that reflect the same set of needs and goals.
- Be used to collect data directly from the users.
- Used alongside user interviews, survey answers, etc., you can also have a user fill in an empathy map themselves. This often reveals aspects of the user that may have remained unsaid or not thought of.

Empathy map

- **Limitations:**
- Empathy mapping may grow with the field; however, it isn't a replacement for journey mapping, scenarios, or user flows—all of which play an equally important role in the design process
- While empathy mapping is a cost-effective and easily transmittable way to create for and cater to the user, it cannot stand alone

Example - Campus Navigation App

- A university is developing a **Campus Navigation Mobile App** to help students, staff, and visitors locate classrooms, labs, libraries, canteens, and offices within a large campus.

Empathy Map for a New Student User

Quadrant	Details
SAYS	"I can't figure out in which building my lab is in." "The directions are too confusing." "I just want something quick and reliable."
THINKS	"If I'm late, the professor might not let me in." "Am I dumb for not finding this classroom?" "This should be as easy as Google Maps."
DOES	Opens app, searches for room number, zooms map in/out. Asks a senior student for directions. Walks in circles near wrong building before finding the right one.
FEELS	Anxious (about being late). Frustrated (when app gives unclear directions). Relieved & Excited (once correct location is found).

Empathy Maps

- **Across the quadrants in Empathy map:**
 - Some of these quadrants may *seem ambiguous* or overlapping- it may be **difficult to distinguish** between *Thinks* and *Feels*
 - if an item may fit into multiple quadrants, just pick one
 - Encounter inconsistencies —seemingly positive actions but negative quotes or emotions coming from the same user
 - 4 quadrants exist only to push our knowledge about users and to ensure we don't leave out any important dimension.

Empathy Maps

- **One User vs. Multiple-Users**
 - Empathy maps can capture one particular user or can reflect an aggregation of multiple users.
 - One-user (individual) empathy maps are *usually based on a user interview* or a user's log from a diary study

Empathy Maps

- **Aggregated Empathy Maps**
 - Represent a user segment, rather than one particular user.
 - Usually created by combining multiple individual empathy maps from users who exhibit similar behaviors who can be grouped into one segment.
 - The aggregated empathy map synthesizes themes seen throughout that user group and can be a first step in the creation of personas.
 - Can be one way to visualize what we know about a persona in an organized, empathetic way

Empathy Maps

- **Aggregated Empathy Maps**
 - Can also become ways to summarize other qualitative data like surveys and field studies
 - Can be used to communicate a persona, instead of the traditional ‘business card’ approach.
 - As more research is gathered about that persona, can add new insights or remove those that have changed or been invalidated.

Empathy map creation

- From the given actions/ observations, classify into 4 quadrants of empathy map (Says, Thinks, Feels ad Does)
 1. Where should I start?
 2. What is the best for me?
 3. Fear
 4. Unsure to trust
 5. Checks the website
 6. Why is this so hard?
 7. I want something reliable
 8. Excited
 9. Postpones big decisions More research

Empathy map creation

- From the given actions/ observations, classify into 4 quadrants of empathy map (Says, Thinks, Feels ad Does)

10. Postpones big decisions More research

11. Lists Pros and Cons

12. Compare products

13. What else am I missing?

14. What size is best?

15. Asks friend

16. What brand do you like?

17. Where should I start?

18. What is the best for me?

19. Fear

Empathy map creation

- From the given actions/ observations, classify into 4 quadrants of empathy map (Says, Thinks, Feels ad Does)

20. Unsure to trust

21. Checks the website

22. Why is this so hard?

23. I want something reliable

24. Excited

25. Postpones big decisions

26. More research

27. Lists Pros and Cons

28. Compare products

Empathy map creation

- From the given actions/ observations, classify into 4 quadrants of empathy map (Says, Thinks, Feels ad Does)

29. What brand do you like?

30. May be this isn't the best

31. Anxious

32. Wasting too much time in searching?

33. Over –whelmed looking at the features

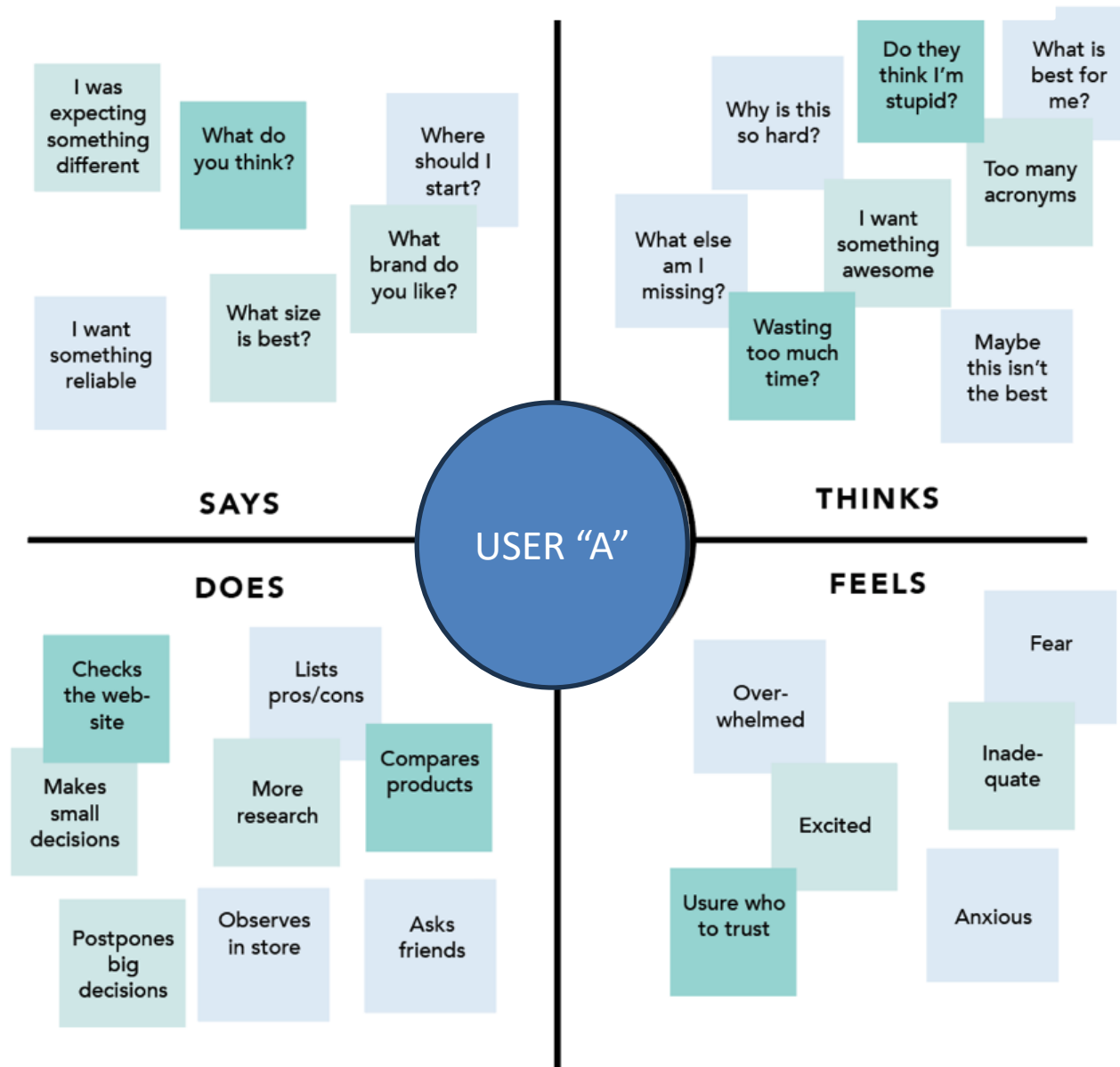
34. Do they think I am stupid?

35. What else am I missing?

36. What size is best?

37. Asks friend

Sample Empathy Map



Wicked problems



Wicked problems

1. Problem is never solved Completely

- User Interfaces, systems evolve with time as technology, requirements, users' expectations may change over time; hence the user experience
- With evolution different issues may arise hence requires different solutions

Wicked problems

2. Every Problem is unique

- **NO** two systems are similar and hence solutions are ***NOT*** similar
- Though UI design may follow a patterns problems lead by

Wicked problems

3. There is no unique solution

- Though the steps and pattern to solve a problem are well researched and practice **NO unique** solution exists
- User Interface design is very creative work and will vary from designer to designer based on the experience, ability, availability of technology, cost, time, interests of various stake holders

Wicked problems

4. Multi-scalar, Multi-causal, interconnected

- Most of the problems are dependent on various factors
- such scope could be local. Regional or global
- more than one factors may affect at a time
- Depends upon number of other factors such as time, circumstances, educational background etc.

Wicked problems

5. Multiple stakeholders with conflicting agendas

- Various stakeholders may expect totally contradicting outcomes and hence satisfying all of them or any of them at all times is not possible

Wicked problems

6. Allowing multidisciplinary activities

- The solution may involve application of various domains such as psychology, technology, accounting and finding out team members having background with such a vast variety of disciplines is difficult

Wicked problems

7. Every wicked problem is connected to other

- The problems are interlinked and if you try to solve one the other may crop up

Wicked problems

8. Every solution ramifies (spreads) throughout the systems

- The suggested solution may have effect on other part of the system
- Selecting appropriate solution is difficult

Wicked problems

9. Solutions are not right/ wrong but better/ worse:

- All the solutions provided will have some merits and limitations in the given constraints and hence no solution will be right or wrong, the degree to which the option can solve the problem may vary and hence the user experience

Wicked problems

10. Can take long time to solutions

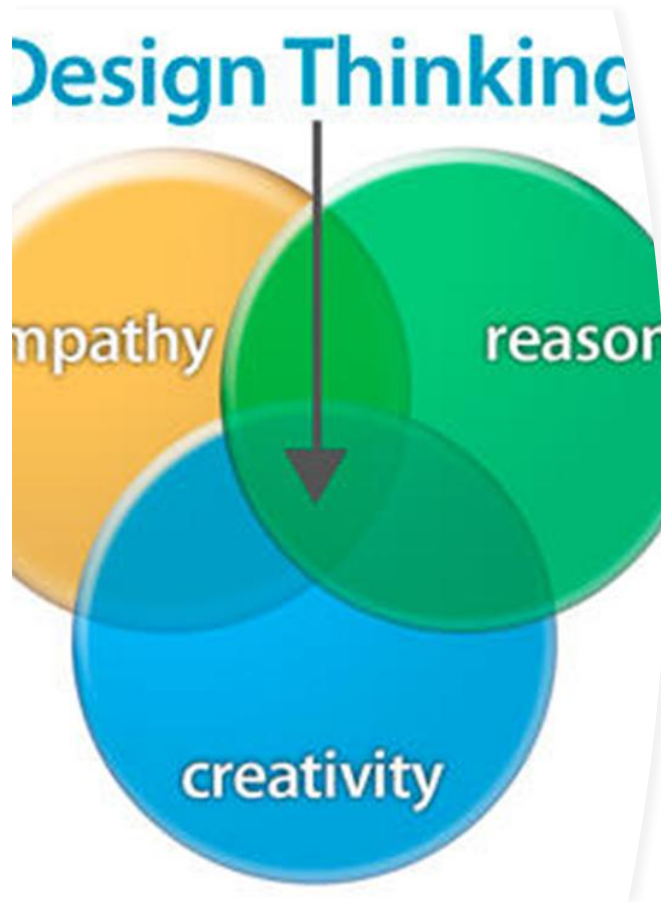
- Solving the problem in the first approach may not be possible
- The evolution of the system through number of iterations may provide the expected results and hence the user satisfaction may increase over time if appropriate steps are taken for improving the system

Wicked problems

- 1. Problem is never solved Completely**
- 2. Every Problem is unique**
- 3. There is no unique solution**
- 4. Multi-scalar, Multi-causal, interconnected**
- 5. Multiple stakeholders with conflicting agendas**
- 6. Allowing multidisciplinary activities**
- 7. Every wicked problem is connected to other**
- 8. Every solution ramifies (spreads) throughout the systems**
- 9. Solutions are not right/ wrong but better/ worse**
- 10. Can take long time to solutions**

Design Thinking

Design Thinking



- “A human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success.”
- Non-linear, iterative process that teams use **to understand users, identify problems, and create solutions**
- Often used to address ill-defined or unknown problems

Design Thinking

5 Stages of the design thinking process:

1. Empathize:
 - Understand the needs of the users by conducting research, observing, and consulting with them.
 - Helps teams put themselves in the users' shoes and understand their experiences, motivations, and problems.
2. Define:
 - Identify the needs and problems.
3. Ideate:
 - Generate ideas via brainstorming or worst possible idea.

Design Thinking

Stages of the design thinking process:

4. Prototype:

- Transform the idea into a tangible solution by building a product that tests ideas and changes.
- Can include mock-ups of every feature and interaction in the final product, allowing teams to check if the idea works and verify the user experience.

5. Test:

- Test the solution.

Wicked problems

- “A social or cultural problem that is difficult or impossible to solve because of its complex and interconnected nature”
 - because
 - often ill-defined
 - information is confusing
 - involve stakeholders with different perspectives
 - have no "right" or "optimal" solution

Wicked problems cannot be solved by the application of standard methods as they demand creative solutions

Ideation.

UX Design Process

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Spotify

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